

A broad study of machine learning and deep learning techniques for diabetic retinopathy based on feature extraction, detection and classification

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ABSTRACT

Diabetic Retinopathy (DR) is a micro vascular disorder that affects the eyes and is a long term effect of Diabetes mellitus. The likelihood to develop diabetic retinopathy is directly proportional to the age and duration of the diabetes, as well as increase in the level of blood glucose level and fluctuation in blood pressure levels. A person who has diabetes has more probability to develop diabetic retinopathy. The ration of people with diabetes started to increase from 285 million in 2010 and will reach up to 439 million in the year of 2030. Out of the total number of people with Diabetic Retinopathy, approximately one-fourth of the people have vision-threatening disease. Earlier detection and classification of Diabetic Retinopathy has to be taken much care in order to sustain a patient's vision. The diabetic Retinopathy may be classified into various stages like Mild non-proliferative retinopathy, Moderate nonproliferative retinopathy, severe nonproliferative Retinopathy and Proliferative diabetic retinopathy. The problem associated with the manual detection of diabetic retinopathy is that the processing time is high, effort consuming and requires an ophthalmologist to examine the eye retinal fundus images. The manual analysis includes Visual Acuity testing, Tonometry and Pupil dilation. The vision lost due to Diabetic retinopathy is sometimes irreparable. Hence there is a need for earlier detection and treatment to reduce the risk of blindness. Hence there are various automated methods of diabetic retinopathy screening that have made good progress using image classification, pattern recognition, and machine learning. The input to the automated image classification model can be the color fundus photography or optical Coherence tomography images.

1. Introduction

Diabetes Retinopathy is a serious eye disease that is caused mainly due to the damage of the blood vessels at the back of the eye. Diabetes is also more likely to develop several other eye conditions like cataracts and open angle Glaucoma. Diabetic retinopathy at its earlier stages will not have any symptoms, but some people may encounter some changes in their vision, difficulty in reading and seeing objects that are far away. After that the blood vessels in the Retina starts to bleed with a gel like fluid. According to statistics 80% of the patients with diabetes who are struggling for a period of 15–20 years suffer from diabetic retinopathy [1]. Diabetic retinopathy due to diabetes is considered as one of the most important cause for vision loss among the people. The severity of

Diabetic retinopathy is correlated with the duration, Blood glucose levels and BP levels. Non proliferative Retinopathy develops initially and causes vision related problems like Micro aneurysms, hemorrhages, Exudates and Macular Edema. Proliferative Retinopathy is likely to develop after Non Proliferative retinopathy and may lead to vitreous hemorrhage and traction retinal detachment. DR can be identified by the presence of various kinds of lesions found on a retinal image. The lesions include Micro-aneurysms, hemorrhages, soft and Hard Exudates [2] (see Figs. 1–4).

Micro-aneurysms: Microaneurysms are the first visible sign of diabetic retinopathy. These seem to be small red dots that appear in the retina. These are surrounded by a ring of yellow lipid.

Hemorrhages: Hemorrhages appear as bigger spots on the retina

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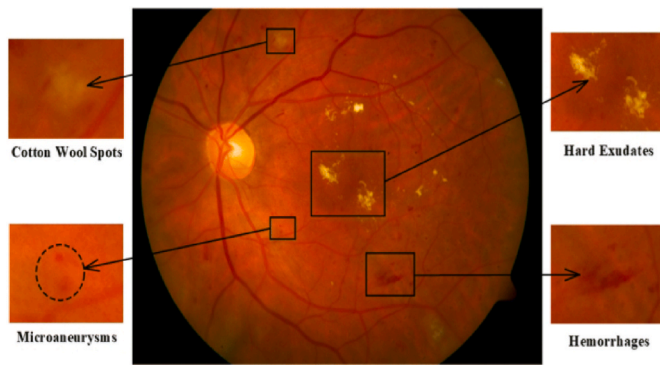


Fig. 1. Soft exudates.

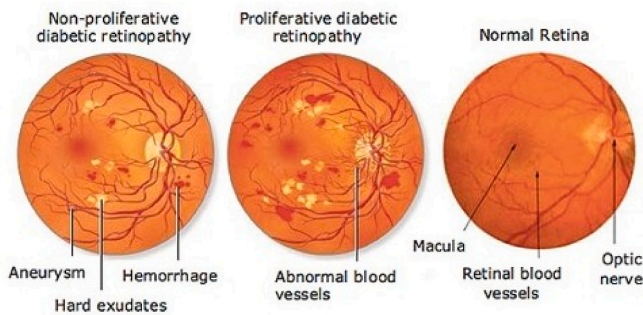


Fig. 2. Types of diabetic retinopathy.

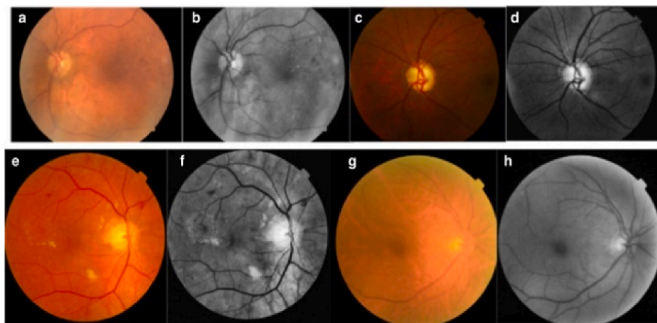


Fig. 3. Eye image before and after contrast enhancement.

and appear like or a splinter along the nerve fiber layer. In the deeper layer they will look like a red dot.

Hard Exudates: Hard exudates are seen as thick yellow spots on the retina which is caused by the leakage of plasma.

Soft Exudates: Cotton wool spots are called soft exudates as they are small and yellow white of gray lesions that appear on the retina. They are less likely to cause vision loss and can resolve within 6–12 weeks.

The manual approach of detecting diabetic retinopathy is the prime cause of misdiagnosis and needs more effort than the automated methods. Diabetic retinopathy may progress through four stages. The initial stages of Diabetic retinopathy will not have any symptoms. The disease habitually grows unseen until it affects vision. Bleeding that comes from abnormal retinal blood vessels can lead to the appearance of "floating" spots. These spots may sometimes go off on their own. But the absence of immediate treatment, may cause bleeding, which increases the risk of permanent vision loss. There are various ways in which diabetic retinopathy can be detected.

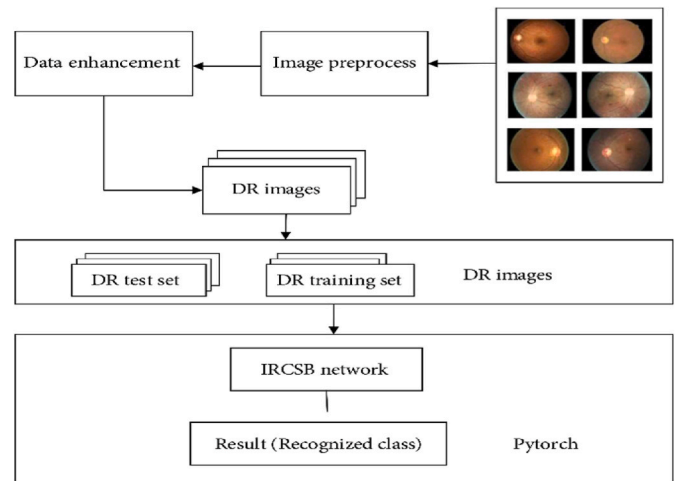


Fig. 4. Architecture diagram of the proposed model.

- Visual Acuity Testing:** A chart is used to measure a person's eye capacity to see at various distances.
- Tonometry:** It is a test that is used to measure the pressure inside the eyes of a person.
- Pupil Dilation:** A Drop is applied on the eye's surface which dilates the pupil, thereby permitting a physician to examine the retina and optic nerve.
- Optical Coherence tomography:** OCT images are used to perform high-resolution and cross-sectional tomographic imaging of the internal structure of various biological systems and materials [3]. OCT functions similar to that of ultrasound B mode imaging which uses light rather than sound. The Image resolutions of 1–15 μm can be obtained, over one order of magnitude which is higher than the conventional ultrasound. Based on the damage caused by diabetic Retinopathy to the eye it can be classified into Proliferative DR and Non Proliferative DR. The basic difference between proliferative and non-proliferative retinopathy is that proliferative retinopathy identifies the presence of neovascularization that causes due to the abnormal blood vessel growth in the retina where non-proliferative diabetic retinopathy identifies the early diabetic retinopathy disease without neovascularization. In the stage of proliferative diabetic retinopathy, abnormal blood vessels and the scar tissues grow on the area of the retina. They are then strongly attached to the back surface of the vitreous gel which then fills the center of the eye. The vitreous then pulls on the scar tissue of the retina, causing the blood vessels to bleed often into the vitreous cavity. This is called as vitreous hemorrhage. This is likely to happen again and again in case of proliferative diabetic retinopathy and thus causes fast and severe loss of vision. These kind of hemorrhages will often be cleared by themselves. The very earliest stage of diabetic retinopathy is called as non-proliferative diabetic retinopathy (NPDR). The early stages of diabetic retinopathy include macular Edema and also the leaking of hard exudates from abnormal blood vessels in the center of the retina, which eventually results in unclear central vision. The limitation of blood supply to the retina occurs, with an increase in

Table 1

Levels of diabetic retinopathy.

Diabetic Retinopathy level and Associated Lesions	
No Diabetic Retinopathy	Absence of lesions
Mild non-proliferative Diabetic retinopathy	Microaneurysm only
Moderate non-proliferative Diabetic retinopathy	More than Micro Aneurysm
Severe non-proliferative DR	No signs of proliferative DR
Proliferative DR	Neovascularization

macular edema. The different levels of diabetic retinopathy are listed in Table 1 [4] (see Table 2).

2. Machine learning and deep learning models for DR detection

Machine learning is a field of Artificial intelligence, which is defined as the ability of a machine to imitate intellectual human behavior. Artificial intelligence based systems are used to perform very complex tasks in such a way as how humans can solve the complex problems. There are a number of machine learning algorithms which plays a vital role in almost all applications like Image recognition, Speech recognition, product recommendations, traffic prediction, self-driving cars, medical diagnosis and so on. Throughout the last decade, artificial intelligence has, into healthcare at a growing pace. Machine learning algorithms works on a lot of stored data. In case of medicine there are huge amount of data to be outlined. Medical professionals used various methods to store and process this data but modern technology goes a step further not only in storing but also in doing the predictions. Based on all the collected data related to the disease, ML algorithms can provide multidimensional solutions which are more accurate than the manual diagnosis. Let's discuss some of the machine learning algorithms that provides some useful insights in the detection and classification of diabetic retinopathy.

- i) **Convolutional Neural Network:** A convolutional neural network or CNN is a deep learning based neural network which is designed for processing structured arrays of data in any forms such as images. These are extensively used in computer vision and has become the center of focus for many applications like Image processing and also has succeeded in Natural language processing for text classification. These works well in finding the patterns in the input images such as lines, circles, gradients or some arbitrary shapes like eyes or faces. A CNN is a feed forward neural network that comes with 20–30 layers. These layers are used to solve the complex problems of disease detection [5–9].

Tools used: Google Net is a deep convolutional neural network which was developed by Google [10]. This can be used to perform various tasks with transfer learning. A simple neural network is a network of neurons with a set of input layers, multiple hidden layers, and an output layer. Each hidden layer is similar to that of a neuron with a set of inputs and weighted outputs, which produces the output through an activation function [11].

- ii) **Support vector machine:** Support vector machine is a supervised machine learning method used for solving various problems like classification, regression, and various other detection problems. SVM can also be used to detect cancerous cells or to predict the driving routes with a well-defined regression model. Some SVMs which are used for particular machine learning problems,

like support vector regression (SVR) is an extension of support vector classification (SVC). SVMs can differ from other classification algorithms in such a way they choose the decision boundary. The decision boundary created by SVMs is called the hyper plane with maximum margin. A simple linear SVM classifier draws a straight line between two classes. All the data points that lie on one side of the line represent one category and that lie on the other side of the line will represent another category. It can solve the problem of classification and regression

- iii) **Naive Bayes Algorithm:** Bayes' theorem is used to calculate the probability of a hypothesis with the prior knowledge. It completely depends on the conditional probability. Bayes' theorem is given by the formula

$$P\left(\frac{q}{r}\right) = \left(P\left(\frac{r}{q}\right) P(q) \right) / P\left(\frac{q}{r}\right) \quad (1)$$

Where, $P\left(\frac{q}{r}\right)$ is called as the Posterior probability that is Probability of hypothesis q on the observed event r. $P\left(\frac{r}{q}\right)$ is called as the Likelihood probability that is Probability of the evidence given that the probability of a hypothesis is true.

- iv) **K nearest Neighbor:** K-nearest neighbor is an algorithm for pattern recognition that make use of the training datasets to find the nearby relatives in the future examples. KNN calculates the distance between the test data and all the training points tries to predict the correct class. Then K number of points are selected that lies close to the test data. The KNN algorithm computes the probability of the test data that belongs to the classes of 'K' training data and the class which holds the highest probability will be selected. In regression, the value is taken as the mean of the 'K' selected training points. The first step is to select the K neighbor. Then next calculate the Euclidean distance of K neighbor. The distance can be calculated using various techniques such as Euclidean, Manhattan and hamming distance. Based on the Euclidean distance that is calculated take the K nearest neighbor. Among these k available neighbor, count the total number of the data points in each category. Finally new data points are assigned to the category for which has the maximum number of the neighbor.
- v) **Decision trees:** A Decision Tree is defined as a supervised machine learning algorithm. It is also a tree based representation of all the possible solutions. All the decisions in the decision tree were made based on some conditions. The decision tree initially starts from the root node and then branches off to the number of solutions just like a tree. The tree also starts from the root and it starts to grow its branches when it grows bigger and bigger. Decision trees can be represented as classification trees or regression trees. Both classification and regression algorithms fall into a category of "classification and regression trees" and are

Table: 2
DR dataset details.

Dataset	Numbers of images	Normal image	Mild DR	Moderate and severe non-proliferative DR	Proliferative DR	Training Sets	Test Sets	Image Size
DiaretDB1	89 images	28 images	6 images	28 images	27 images	28 images	61 images	1500 × 1152 pixels
DRIVE	40 images	34 images	6 images	–	–	20 images	20 images	565 × 584 pixels
Messidor	1200 images	–	–	–	–	–	–	Different resolution
Messidor-2	1748 images	–	–	–	–	–	–	Different resolution
STARE	20 images	10 images	–	–	–	–	–	700 × 605 pixels
CHASE DB1	28 images	–	–	–	–	–	–	1280 × 960 pixels
IDRiD	516 images	–	–	–	–	413 images	103 images	4288 × 2848 pixels

sometimes referred to as CART. The CART algorithms are used to solve the classification and prediction problems.

3. Review of previous work

Mahmoud Elgaf et al. extracted the 3D features from all the retinal layers that include the first-order reflectivity and also the 3D thickness of the OCT layers. The Back propagation neural networks were used to classify the Optical Coherence Tomography images which gives an accuracy of 96.81% that uses leave-one-subject out (LOSO) cross-validation method [1]. Their work performed the comparison of the proposed system with the most popular Deep Learning algorithms like Google Net and Resnet-50. Transfer learning technique has been applied to avoid over fitting over these Deep learning algorithms. Google Net gas achieved the best accuracy of 93.98% by using 5 folds and 94.43% by using 10 folds [1].

NajibUllah et al. proposed a deep neural network for the proposed work. A genetic algorithm-based feature selection approach has been used for feature selection. They applied five advanced CNN architectures for extracting the features from the fundus images. The CNN architectures used are Alex Net, NASNet-Large, VGG-19, Inception V3, and Shuffle Net. Then genetic algorithms are applied for feature selection and for ranking the features into two categories such as high rank and lower rank. The non-optimal feature attributes that are obtained from training data sets are dropped. Support vector machine classification model is used to classify the diabetic retinopathy detection and classification. Finally the performance of the proposed model is evaluated using various metrics like accuracy, specificity, sensitivity, precision, recall, and F1 score. Then the proposed model is finally tested with different datasets like the Kaggle dataset, a customized dataset, and an enhanced custom dataset with 97.9%, 94.76%, and 96.4% accuracy, respectively [2]. The metrics that are used to measure the performance of the system are accuracy, recall, specificity and precision.

$$\text{Accuracy} = (TN + TP) / (FP + TN + TP + FN) \quad (2)$$

$$\text{Recall} = TP / (FN + TP) \quad (3)$$

$$\text{Specificity} = TN / (FP + TN) \quad (4)$$

$$\text{Precision} = TP / (FP + TP) \quad (5)$$

Angel Ayala et al. implemented a CNN to process a fundus image for recognizing the structure of the eyeball and determined the existence of diabetic retinopathy. Transfer-learning methodology is used to optimize the parameters where an image is mapped with the corresponding label. The medical fundus images are divided into training and test images and a severity scale found in the eyeball as labels. The severity scale categorizes the images into five classes that ranges from a healthy eyeball to a proliferative diabetic retinopathy eye. The final stage is probably a blind stage. The proposed methodology gives an accuracy of 97.78% [3]. Yun et al. proposed a method to classify the retinal fundus images into four different stages namely normal, moderate, severe, and proliferative Diabetic retinopathy. The first step is to preprocess the image. The images were preprocessed using some morphological operations with disc and diamond structuring elements. Considering the perimeter and area of the pixels in the RGB channels six features were obtained. For classification, a single-layer feed-forward neural network was used which consists of eight units in the hidden layer. Six units were used for each feature value as inputs and four units as output, one for each severity of Diabetic Retinopathy [12].

Hoover et al. proposed an automated method that can locate the blood vessel using the local and the global vessel features [13].

Murat canayaz proposed an automatic image evaluation method which is required for the diagnosis of diabetic retinopathy. A new approach has been used for feature selection from the fundus images using wrapper methods. Feature selection is the most prominent process

in diabetic retinopathy detection. The wrapper methods are used to select the most significant and relevant features from the set of huge features given in the dataset. They evaluate on a particular machine learning algorithm to find the optimal features from the available features. The wrapper algorithms selected for the proposed approach includes gravity search algorithm, gray wolf algorithm, equilibrium optimizer and binary bat algorithm. The proposed method has obtained an accuracy of 96.32 and 0.98 kappa [14].

Rubinasarki et al. proposed an automatic classification system based on two scenarios such as multi class diabetic eye disease and mild multi class diabetic eye disease. The detection is done using two pretrained CNN models on the ImageNet. The proposed method also focuses on various performance improvement techniques such as contrast enhancement and fine tune optimization. The model achieved an accuracy of 88.3% for multiclass diabetic eye disease and 85.95% for mild multi class diabetic eye disease [15]. The fig below depicts the original eye image and image after contrast Enhancement.

Yue Miao et al. proposed a Hybrid attention mechanism that was combined with residual CNN model algorithm which can clearly extract the features of very small lesions better than the general extraction methods. Initially a Multiscale deep learning based network model with hybrid attention was used to extract the high level features of the image, then different types of samples are sampled by sampling algorithm, and finally the extracted features are enhanced based on the spatial attention and channel attention. The proposed algorithm has the tendency to learn more features and also has improved the classification performance of diabetic retina which shows an accuracy of 93.8 % [16].

Wejdan L. Alyoubi et al. proposed a system that classifies the DR retina Images into five stages such as No DR, Mild DR, Moderate DR, Severe DR and Proliferative DR. The system also localizes the affected lesions on retinal surface of the eye. The proposed system has been implemented using two models. The first model uses the image as input to the CNN model and classifies the image into one of the five classes as stated earlier and achieved an accuracy of 88.6%. The second model uses an adopted YOLOv3 model for the detection and localization of the DR lesions. This achieves a 0.216 mAP in the localization of the lesion [17].

Nagaraja Gundluru et al. designed a deep learning model where PCA is used for dimensionality Reduction and Harris Hawks Optimization Algorithm is used for optimizing the classification and feature Extraction process [18]. Shahnawaz Ayoub et al. used deep learning techniques to detect the blind the spots in eyes. The proposed method used an enhanced convolutional neural network (ECNN) which uses a genetic algorithm. The noise is filtered using Gaussian filter and the features are extracted using the genetic algorithm [19].

Malik Bader Alazzam et al. focused his research on the study of the patients who are suspected for diabetic retinopathy. OPF and Restricted Boltzmann machines are used to analyze and classify the retinal fundus images. The models extracted five hundred and one thousand characteristics for the classification of the disease after training for the recognition of the disease. The experiment was conducted with fifteen different series, with 30 cycles as the repetition rate. The RBM yields the highest diagnostic accuracy of 89.47 and OPF has a better auto detection system with 100% specificity. Hence the proposed model for the detection of diabetic retinopathy performs better in terms of accuracy and sensitivity [20].

Asif Irshad Khan et al. proposed a technique based on deep learning for the classification of retinal images to identify diabetic retinopathy. A binary classifier was used to classify the retinal image whether the image is being malicious or normal. The binary classifier algorithm has given an accuracy of 96.68% and validation accuracy of 66.82% with the dense neural network [21]. For imbalanced diabetic retinopathy data distributions a new category attention block was used that identifies more discriminative region of interest features for each grades of diabetic retinopathy [22].

Deep learning is a subdivision of machine learning that has the

ability to solve high potential problems in different fields. The Deep learning models usually consists of multilayered neural network algorithm which involves extracting the common features in very complex datasets [23]. Deep learning tools also have proved the efficiency and effectiveness of solving various tasks which includes natural language, understanding, content classification, sentiment analysis, question answering and language translation. Thus Deep learning plays a vital role in the detection of various diseases [24]. The main important strategy in deep learning include transfer learning [25–28].

M.Imran et al. proposed an entropy-based enhancement technique which is used for refining the features of the images on the dataset which is not balanced. This method has been obtained from an enhanced classification using a hybrid neural network and low computational cost. In addition, to the proposed model a modified deer hunting optimization algorithm is used for extracting optimal features for enhanced classification accuracy [11,29].

V. Purna Chandra Reddy et al. created a graph learning-based graph convolutional network (GCN) implementation for the more accurate categorization of combined DR-DME. To extract the deep features of a single DME, DR, and joint DR-DME, a hybrid GCN (HGCN) with relation aware channel-spatial attention (RACSA) model is initially created. Additionally, a cutting-edge bio-optimization strategy called the modified deer hunting optimization algorithm (MDHOA) is used as the best feature extraction method. Finally, the classification of distinct DME, DR, and joint DR-DME disorders is performed using HGCN as a classifier. The detailed simulations performed on the IDRid dataset demonstrate that the proposed OHGCNet outperformed the traditional approaches, with improvements in classification accuracy of 5.11%, 3.88%, and 5.47% for DME, DR, and joint DR-DME [30].

A. Tayalet al presented a model that automatically identifies the diabetic macular edema, drusen, choroidal neovascularization, and normal pictures in optical coherence tomography (OCT) scans of the human eye, this study proposes a diagnostic tool based on a deep-learning framework for four-class classification of ocular illnesses. The suggested framework makes use of OCT retinal images and analyses with three different convolution neural network (CNN) models (five, seven, and nine layers) to recognize the various retinal layers and extract useful information from them, watch for any new deviations, and forecast the various eye deformities. The framework makes use of OCT retinal images that have been enhanced for contrast, noise removal, contour-based edge recognition, and retinal layer extraction. Three alternative CNN models (of five, seven, and nine layers) are used to analyze this image collection [31]. CNN finds its place not only in eye disease classification but also in the fracture detection [32].

4. Dataset for retinal images

A dataset is a structured collection of data which acts as an effective tool for analyzing and tracking important information associated with a work. The datasets are used to provide valuable information about the severity of diabetic retinopathy which makes it seamless for the development and analysis of image evaluation algorithms for the earlier detection of diabetic retinopathy. There exists many publicly available datasets for the retina for the detection and classification of diabetic retinopathy. These are mostly used to train, test and validate the models. The images may be categorized as fundus images and Optical Coherence Tomography Images. The OCT images are represented as three dimensional images taken using the low coherence light and provides useful information about the retinal structure and thickness. The fundus images are represented as the two dimensional images of the retina that has been taken using the reflected light [33].

STARE: The structured analysis of Retina database was generated by scanning and digitizing the existing photographs of the retinal images. These images are captured by a field view of 35° camera and have the resolution of 700x605 pixels. It consists of 400 raw images.

MESSIDOR: Messidor stands for Methods to evaluate segmentation

and indexing techniques in the field of retinal ophthalmology. The Messidor database captured the fundus images with a 3CCD video camera that has a field of view of 45°. The images are of different resolutions such as 1140 X 960, 2240 X 1488, and 2304 X 1536 pixels. It consists of 1200 retinal images in the dataset.

MESSIDOR-2: Messidor –2 is an extended dataset of messidor. The Messidor-2 dataset is a collection of retinal images for diabetic retinopathy with each eye consisting of two specifications for the macula-centered eye fundus images. Messidor original consists of all image pairs from the original Messidor dataset. The Eye fundus were imaged, using a Topcon TRC NW6 non-mydratic fundus camera. Messidor-Extension contains 345 examinations. Messidor-2 contains 1748 images.

DIARETDBO AND DIARETDBI: Diabetic Retinopathy Database Calibration level 0 is a database consisting of 130 retinal images which are taken using a digital fundus camera that has a field of view of 50°. 130 images are taken out of which 20 are healthy and the remaining 110 images contains the symptoms of DR. Diabetic Retinopathy Database Calibration level 1 database has 89 retinal fundus images. Out of the available 89 images, 84 retinal images contain the signs of mild non-proliferative diabetic retinopathy (NPDR), and the remaining 5 are healthy. The images are captured using the field view of 50°. The image resolution found in both DIARETDB1 and DIARETDB0 database is 1500 X 1152 [34].

REVIEW: Retinal Vessel Image set for Estimation of Widths database was developed in the University of Lincoln, UK by a group of researchers. This database has been developed explicitly to act as a reference for various vessel segmentation algorithms. It contains the details about the vessel width of 16 images which are included in this database. The database is divided into four segments, i) High-resolution image set, ii) vascular disease image sets, iii) central light reflex image set and the iv) kick point's image sets.

IDRid: Indian Diabetic Retinopathy Image Dataset (IDRid) database is used to estimate the performance of algorithms that are developed for the automatic detection and classification of Diabetic retinopathy and DME using the retinal fundus images and OCT images. The database is a collection of 516 images where 413 images are used for training and 103 images are used for testing. The manually marked OD center and the fovea location are available for the images in the dataset.

Images were captured using a Kowa VX-10 alpha digital fundus camera with a field of view of 50°, and all the images are centered near to the macula. The images have a resolution of 4288 X 2848 pixels [35, 23]. Retinal fundus imaging is non-intrusive and painless mean to screen the retina [9,21]. The IDRid database fundus images were acquired from an Eye Clinic which is located in Nanded, (M.S.), India. The Retinal images of humans affected by diabetes were captured with 39 mm distance between lenses. The eyes are examined using non-intrusive fundus camera having xenon flash lamp [36].

EyePACS: EyePacs is a public freely available web-based application digital image system. This system is mainly used for storing and managing the retinal images of the patients. EyePacs have provided the public a large dataset of retinal images from diabetic detection and classification problems. The dataset can be downloaded from Kaggle. This database consists of 35,126 images which are acquired with a variety of fundus cameras [35].

ORIGA: ORIGA stands for Online Retinal Fundus Image Dataset for Glaucoma Analysis and Research ORIGA database contains 650 images that are acquired using the Singapore Malay Eye Study (SiMES). The ORIGA dataset includes 168 glaucomatous and 482 non-glaucoma images.

CHASE: Child Heart and Health Study in England (CHASE) database is a dataset for retinal vessel reference which was acquired from the multi-ethnic school children. The CHASE database is observed to contain 28 retinal images that was taken using a hand-held NM-200-D fundus camera. The images in the CHASE database have a resolution of 999 X 960. Apart from these datasets kaggle is also of the most widely used public dataset for retinal eye disease detection [37,38].

RIDB is a recreation information database which is comprised of Retinal fundus images that are used to extract the retinal features for the detection and classification of diabetic retinopathy from retinal images. The database is also significant in the analysis of retinal features that aids the researchers to carry out the research based on the retinal image classification [39].

The RFMiD is an open source publicly available retinal images dataset that consists of around 3200 images. The images in the dataset are divided into two categories which are the Screening of retinal images into normal and abnormal categories and Classification of retinal images into 45 different categories. The dataset is split into 3 groups: 60% of the image for training, 20% of the image for evaluation, and the remaining 20% of the Images for Testing. The main objective of RFMiD dataset is to provide the details of different kinds of diseases that appear in usual clinical practice [40,41,42].

5. Feature extraction techniques

Shanthi et al. has proposed the modified Alexnet architecture based on CNN for DR detection and classification. The Classification was performed over the MESSIDOR dataset and has produced an accuracy of 96%. The U-Net residual network with the pre-trained ResNet34 model was presented for the segmentation of Diabetic Retinopathy lesions. Alexnet architecture gives the accuracy of 96.6% for classification on grade 0 and 96.2% on grade 1, 95.6% on grade 2 and 96.6% on grade 3 [43]. In a proposed work the input images are preprocessed and converted into gray scale images. The RGB Segmentation is mainly used for dividing the images into various segments based on their characteristics for extracting the curves and lines of the image [44]. The feature extraction using ULBPEZ method resizes the fundus images to a fixed size of 512x512 pixels. The images are divided into various smaller blocks where each block is processed for the uniform Local Binary pattern feature extraction stage. The local binary pattern coefficients are computed based on the circular neighborhood with the radius. The uniform LBP features that are extracted from each block are concatenated with the previously calculated LBP values of the previous block [45]. Hralick computed 24 statistical features that are different from the normalized GLCM matrix. These Statistical features calculates the important part of the local information and spatial features that are located in the image. Out of the 24 features only 5 features are used to extract the required information which was selected after a number of trial combinations [46,47]. T Anuradha et al has proposed a work which best illustrates the need for CNN for extracting the features. It focused on the existing feature extraction methods and also discussed about the importance of CNN for feature extraction [48].

6. Conclusion

The early identification and classification of diabetic retinopathy is very much important for the prevention of loss of vision [49]. Thus the comprehensive study of different machine learning and deep learning algorithms helps analyze the different kinds of algorithms for diabetic retinopathy detection. Diabetic Retinopathy is one of the complications of diabetes. Effective and automated diagnosis to the degree of Diabetic Retinopathy lesions have an important clinical significance. The Early diagnosis allows for early treatment, which is crucial because early detection can effectively prevent visual impairment [50]. The main issue addressed in the reviewed studies is the manual diagnosis that has to occur after screening, which is typically a lengthy process prone to ophthalmologists' bias. Moreover, dataset limitations restrict fundus image variations that can be used in the assessment of indicators [51].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

No data was used for the research described in the article.

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